Rupesh Radhakrishna Nipanikar

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Education

Swami Vivekanand Institute of Polytechnic (SVIP) **2020 – 2023**
B.Tech in Computer Science and Engineering CGPA: 7.97

Tssm's Bhivarabai Sawant College of Engineering and Research **2023 – 2026**
B.Tech in Computer Science and Engineering CGPA: 7.97

Skills

- **Programming Languages:** Java, Python, MySQL
- **Technical Skills:** DSA, Data Analytics, Full Stack Development (Frontend, Backend, Database, Deployemnt), Cloud Computing (AWS)
- **Tools & Visualization:** Power BI, DAX, RAG, LangChain, Scikit-learn, Pandas, NumPy, Matplotlib

Projects

AI Animation Generator using Gemini and Manim **Jan 2026**
LLM Application Project

- Developed an AI-powered text-to-animation system that converts natural language prompts into executable Manim animations using Google's Gemini LLM and LangChain.
- Engineered structured prompt templates and Pydantic-based output validation to generate reliable, error-free Python animation scripts.
- Automated the end-to-end workflow including code generation, script execution, video rendering, and playback within Google Colab.
- Implemented layout and rendering constraints to ensure visually consistent, non-overlapping animations while adhering to Manim best practices.

Pneumonia Detection Web Application — Chest Insight Project **Apr 2025 – May 2025**
Healthcare AI Project

- Created a convolutional neural network (CNN) model using pre-trained VGG16 architecture on 1000+ chest X-ray images
- Developed a Streamlit-based web app that allows users to upload X-rays and receive real-time pneumonia detection results
- Achieved classification accuracy of 90%+ on validation datasets, demonstrating strong diagnostic potential
- Designed a user-friendly interface for doctors and patients, enabling seamless image upload and presenting a definitive pneumonia detection result.

Customer Churn Prediction — Machine Learning, Streamlit **Feb 2025**
Predictive Modeling Project

- Performed data processing, cleaning, and feature engineering using pandas and NumPy on 5000+ customer datasets
- Trained and evaluated multiple machine learning models including Logistic Regression, SVM and Random Forest using scikit-learn; Logistic Regression achieved the best prediction accuracy of 89.5%
- Built interactive churn insights visualizations with matplotlib and reduced prediction latency for faster decision-making
- Developed a Streamlit-based web application improving business planning speed through real-time insights

Certifications

- Full Stack Development (AI Engineering)

Leadership / Achievements

- NSS Coordinator: Coordinating student volunteers, Managed event coordination, volunteer participation, and communication to ensure smooth execution of NSS initiatives.